

Riley Beenders

CONTROLS AUTOMATION | MOTION CONTROL | SOFTWARE-INTEGRATED SYSTEMS

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PROFESSIONAL SUMMARY

Controls and electromechanical R&D leader with 10+ years of multidisciplinary engineering experience, including 6+ years delivering industrial automation, motion-control, and machine-integration systems. Leads architecture, risk analysis, software/hardware integration, validation, troubleshooting, and technical documentation from concept through commercialization.

CORE EXPERTISE

Controls & Motion: Control-system architecture; closed-loop servo and linear-motor systems; Allen-Bradley PLCs; CAN bus; Modbus; I/O and sensor integration; tuning and calibration

Software & Tools: C/C++ (working knowledge); LabVIEW; MATLAB, Simulink, and Simscape; Python; Visual Studio Code (IDE); Git/GitHub

Safety & Delivery: Safety, operational, and software risk analysis; validation workflows; debugging; technical writing; electrical schematics and BOMs; operator training; cross-functional leadership

PROFESSIONAL EXPERIENCE

Proteor / Filament Innovations | Coplay / Whitehall, PA

May 2020 - Present

Lead R&D Engineer (2025 - Present) | Chief Technology Officer (2023 - 2025) | Progressive engineering roles (2020 - 2023)

- Led cross-functional product and controls programs from concept through commercialization, with hands-on ownership of mechanical, electrical, firmware, and software architecture, integration, troubleshooting, and documentation.
- Standardized linear-motor architecture across POSEIDON and ICARUS platforms and validated system performance at 4-micron precision, up to 4 m/s, and a 30 kg moving mass.
- Performed and documented safety, operational, and software risk analyses; implemented power-failure braking and other mitigations supporting EU and U.S. compliance.
- Integrated and debugged modular CAN-bus controls and independently testable electrical enclosures, reducing wiring complexity and enabling rapid installation and subsystem validation.
- Built a LabVIEW/Modbus metrology workflow collecting nine readings every 100 ms from multi-axis laser sensors for automated tolerance validation, quality control, and customer reporting.
- Led machine integration for an operator-facing remote control and diagnostics interface with live monitoring, alerts, and production metrics; also directed firmware tuning, calibration, operator training, and post-delivery support.

ADDITIONAL RELEVANT EXPERIENCE

Blue Mountain Ski Resort - Ski Lift Operator | 2019 - 2021 | Performed motor/control safety checks on passenger lifts and supported maintenance during breakdowns while assisting thousands of guests.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and design/manufacture 3D-printing hardware accessories.

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Automated 2,500+ device deployments with an Arduino-based setup tool and deployed 750+ wireless access points.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: MathWorks MATLAB, Simulink, and Simscape Onramps (2026); Lean Six Sigma Yellow Belt (2025); Penn State Certificate in Engineering Design and Tools (2023)